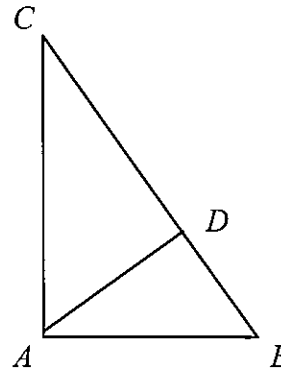


8. If three real geometric means are inserted between 15 and $\frac{5}{27}$, find the value of the middle term. Express your answer as an improper fraction reduced to lowest terms.

9. In the diagram $\triangle CAB$ is a right triangle with $\angle CAB = 90^\circ$. Point D is on $\overline{BC}$ such that $\overline{AD} \perp \overline{BC}$. Let the lengths of $\triangle CAB$ be a , b , and c with these lengths respectively opposite the angles with vertices at A , B , and C . If $b \sin C (b \cos C + c \cos B) = 41$, and if the length of $\overline{AD}$ is an integer, find the largest possible value of AD .



10. The roots of $x^3 + kx^2 - 59x + 420 = 0$ are 5, -7 , and 12. Find the value of k .
11. When $(2x - 3)^8$ is expanded and completely simplified, find the numerical coefficient of x^5 .
12. The first term of an infinite geometric sequence of real terms is 225, and the fourth term of this geometric sequence is 14.4. Find the sum of the terms of this infinite geometric sequence.
13. The sum of the terms of an arithmetic sequence with 13 terms and common difference of 3 is found to be 286. Find the thirteenth term of this sequence.
14. Let there be k objects, each of weight w . When these objects are weighed in pairs, the sum of the weights of all possible pairs is 770. When these objects are weighed in groups of four, the sum of the weights of all possible groups of four is 9240. Find the value of $(k + w)$.